

SUFFERING IN THE WILD

Wild animal suffering is the suffering experienced by wild animals as a result of natural processes such as disease, injury, parasitism, starvation, dehydration, weather, natural disasters, predation, and psychological stress. The topic concerns the extent of these harms, their [moral status](#), and whether humans should respond to them.

Writers on the subject have discussed the large numbers of wild animals affected by these conditions. Some arguments draw on [evolutionary processes](#), including [natural selection](#), high reproductive rates, and high juvenile mortality, to argue that suffering in the wild may be widespread.

Religious, philosophical, and literary works have long addressed suffering in nature. In religious thought, it has been discussed in relation to the [problem of evil](#) and [theodicy](#). From the eighteenth century onward, writers commented on predation, famine, disease, and violent death among wild animals. In the late twentieth and early twenty-first centuries, the subject became a more explicit topic in [animal ethics](#) and [environmental ethics](#).

A central debate concerns whether humans should intervene in nature to reduce suffering. Arguments for intervention are usually based on [animal welfare](#), [animal rights](#), or [anti-speciesist](#) views, and may also note that humans already alter nature for human purposes. Arguments against intervention often appeal to uncertainty about ecological effects, the value of natural processes, wilderness, or wild animal autonomy, and some compare large-scale intervention to colonial rule or paternalism.

Existing and proposed responses include rescue and rehabilitation, vaccination, contraception, feeding and shelter during emergencies, habitat management, and technological proposals aimed at reducing specific harms. The topic has also been discussed in relation to [rewilding](#), climate change, [welfare biology](#), and the possibility of spreading wild animal suffering beyond Earth.

Extent of suffering in nature

Sources of harm

Disease

Wild animals may suffer from diseases that circulate similarly to human colds and flus, as well as [epizootics](#), which are analogous to human [epidemics](#).^[1] Examples include [chronic wasting disease](#), [white-nose syndrome](#), [devil facial tumour disease](#), [Newcastle disease](#), [myxomatosis](#), [viral haemorrhagic disease](#), and [chytridiomycosis](#).^{[1][2]} Diseases and parasitism may produce symptoms such as listlessness, ulcers, pneumonia, starvation, and other harms before death.^[3]

Poor health may increase infection risk, which can further worsen health and increase later vulnerability.^[4] The [terminal investment hypothesis](#) holds that infection can lead some animals to use remaining resources for reproduction.^[5]

Injury

Wild animals can be injured by predation, [intraspecific competition](#), accidents, self-amputation, [molting](#), extreme weather, and natural disasters. Injuries can be painful and may affect movement, feeding, drinking, escaping predators, and interactions with other members of the same

species. Injuries can also increase vulnerability to disease, parasites, starvation, dehydration, and further attacks.^[6]

Parasitism

Many wild animals, particularly larger ones, are infected with parasites. Parasites may affect hosts by redirecting resources, damaging tissue, and increasing vulnerability to predation.^[1] They may also reduce host movement, reproduction, and survival.^[7] Parasites can alter host bodies or behavior, as in limb malformations in amphibians caused by *Ribeiroia ondatrae* or behavioral manipulation in some insects.^{[8][9]} One meta-study found mortality to be 2.65 times higher in animals affected by parasites than in unaffected animals.^[10]

Parasitoids, including some worms, wasps, beetles, and flies, differ from parasites because they kill their hosts.^[11] Their larvae may feed on host organs and fluids until the host dies.^[12]

Starvation and malnutrition

Starvation and malnutrition particularly affect young, old, sick, and weak animals. They can be caused by injury, disease, poor teeth, and environmental conditions, with winter associated with increased risk.^[13] Food availability limits wild animal populations, and deaths from starvation have been described as prolonged and distressing.^{[14]:67} Fish larvae may also experience hydrodynamic starvation, in which fluid motion limits feeding ability; mortality may exceed 99%.^[15]

Dehydration

Drought can cause animals to die of thirst and can increase predation risk when animals leave hiding places to seek water. Dehydration may also combine with starvation, and diseases such as chytridiomycosis can increase dehydration risk.^{[16][17]}

Weather Conditions

Weather affects wild animal health and survival. Heavy snow, flooding, drought, heat, and cold can directly harm animals and can increase risks such as starvation and disease.^[19] Extreme weather can kill animals by destroying habitats or killing them directly.^[20] Hailstorms have killed large numbers of birds.^[21] Extreme heat and drought can dry out vegetation and water sources, and hot water can make it harder for fish to breathe.^{[22][23]} Climate change can increase heat stress and reduce available water sources.^[24] A study on cottontail rabbits found that only 32% survived winter.^[18]